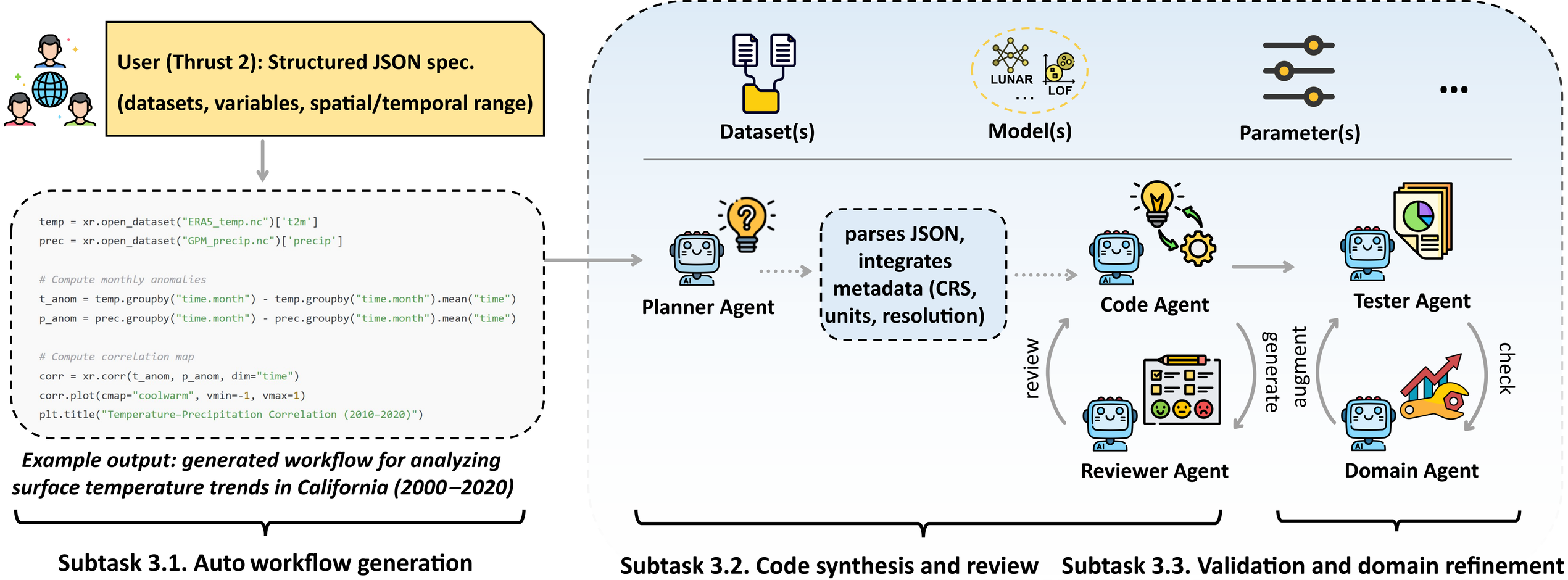
User (Thrust 2): Structured JSON spec.
(datasets, variables, spatial/temporal range)
temp = xr.open_dataset("ERA5_temp.nc")['t2m']
prec = xr.open_dataset("GPM_precip.nc")['precip']
# Compute monthly anomalies
t_anom = temp.groupby("time.month") - temp.groupby("time.month").mean("time")
p_anom = prec.groupby("time.month") - prec.groupby("time.month").mean("time")
# Compute correlation map
corr = xr.corr(t_anom, p_anom, dim="time")
corr.plot(cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Temperature-Precipitation Correlation (2010-2020)")
Example output: generated workflow for analyzing surface temperature trends in California (2000–2020)
Subtask 3.1. Auto workflow generation
Dataset(s)
LUNAR
LOF
Model(s)
Parameter(s)
Planner Agent
parses JSON, integrates metadata (CRS, units, resolution)
Code Agent
Tester Agent
review
generate
augment
check
Reviewer Agent
Domain Agent
Subtask 3.2. Code synthesis and review
Subtask 3.3. Validation and domain refinement